Problem 12.526

Let A be a 7×7 matrix such that

$$2A^2 - A^4 = I_7 \tag{1}$$

where I_7 is the identity matrix. If A has two distinct eigenvalues and each eigenvalue has geometric multiplicity 3, then the total number of nonzero entries in the Jordan canonical form of A equals:

(A) 6 (B) 7 (C) 8 (D) 9

Given Data

Symbol	Meaning	Value
A	Square matrix	7×7
Equation	Polynomial relation	$2A^2 - A^4 = I_7$
Eigenvalues	Distinct eigenvalues of A	Two distinct ($\lambda = \pm 1$)
Geometric multiplicity (GM)	$\dim(\ker(A - \lambda I))$	3 for each eigenvalue
Total GM	Sum over both eigenvalues	$3 + 3 = 6$

Theory

1. Geometric Multiplicity

For a square matrix A and eigenvalue λ,

$$\text{GM}(\lambda) = \dim(\ker(A - \lambda I)). \tag{2}$$

It represents the number of linearly independent eigenvectors for λ.

If the **algebraic multiplicity (AM)** (how many times λ repeats in the characteristic polynomial) equals the GM, then A is diagonalizable. Otherwise, A is **defective** and has Jordan blocks larger than 1×1.

2. Jordan Canonical Form

Every square matrix A is similar to a matrix J such that

$$A = PJP^{-1} \tag{3}$$

where J is the **Jordan canonical form** and P is an invertible matrix whose columns are eigenvectors and generalized eigenvectors.

A Jordan block of size k corresponding to an eigenvalue λ is:

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & \lambda & 1 \\ 0 & \cdots & \cdots & 0 & \lambda \end{pmatrix}. \tag{4}$$

The 1 above the diagonal connects a **generalized eigenvector** to a true eigenvector. For example:

$$\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \tag{5}$$

represents one eigenvector $\mathbf{v_1}$ satisfying $(A-\lambda I)\mathbf{v_1} = 0$, and one generalized eigenvector $\mathbf{v_2}$ satisfying $(A - \lambda I)\mathbf{v_2} = \mathbf{v_1}$.

3. How Jordan Block Sizes are Determined Using Null Spaces

We define

$$N_k(\lambda) = \ker((A - \lambda I)^k). \tag{6}$$

The spaces satisfy

$$N_1(\lambda) \subseteq N_2(\lambda) \subseteq N_3(\lambda) \subseteq \cdots \tag{7}$$

and their dimensions reveal Jordan block structure:

Quantity	**Meaning**
$\dim N_1(\lambda)$	Number of Jordan blocks (= GM)
$\dim N_2(\lambda) - \dim N_1(\lambda)$	Number of blocks of size ≥ 2
$\dim N_3(\lambda) - \dim N_2(\lambda)$	Number of blocks of size ≥ 3

Thus, computing $\ker((A - \lambda I)^k)$ for successive k directly determines how many blocks of each size exist.

Solution

Step 1: Simplify the given polynomial

$$2A^2 - A^4 = I_7 \Rightarrow A^4 - 2A^2 + I = 0 \tag{8}$$
$$\Rightarrow (A^2 - I)^2 = 0 \tag{9}$$

Step 2: Find eigenvalues

Let λ be an eigenvalue of A.

$$(\lambda^2 - 1)^2 = 0 \Rightarrow \lambda^2 = 1 \tag{10}$$
$$\Rightarrow \lambda = \pm 1 \tag{11}$$

So, A has two distinct eigenvalues: $+1$ and -1.

Step 3: Use geometric multiplicity data

Given:

$$\text{GM}(+1) = 3, \quad \text{GM}(-1) = 3. \tag{12}$$

So total eigenvectors $= 6$. Since A is 7×7, one dimension is missing — corresponding to one generalized eigenvector.

Hence, there must be one Jordan block of size 2, and five of size 1.

Step 4: Construct the Jordan form

Assuming the 2×2 block belongs to $\lambda = +1$:

$$J = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{pmatrix}. \tag{13}$$

Step 5: Count nonzero entries

Each 1×1 block contributes one nonzero (the eigenvalue). The 2×2 block contributes three nonzeros (two diagonals and one above it):

$$\text{Total nonzeros} = 5 \times 1 + 3 = 8 \tag{14}$$

$$\boxed{\text{Answer: 8}}$$

Conclusion

The null spaces $\ker((A-\lambda I)^k)$ reveal how large the Jordan blocks are for each eigenvalue. Each difference in their dimensions gives the count of blocks of size $\geq k$.

Since only one eigenvector is missing from full diagonalizability, exactly one 2×2 block exists.

$$\boxed{\text{Total nonzero entries in Jordan canonical form of } A = 8.}$$